

INTERVIEW QUESTION BANK

Associate Director – AI Architect

Enterprise AI Practice · 3-Round Process

Company	Big
Role	Associate Director – AI Architect (IC)
Geography Supported	United States
Total Questions	45 questions across 3 rounds
Reporting To	Director / Senior Director

Interviewer Usage Guide

This question bank is structured around BigComp's 3-round process. Each round has a distinct purpose; interviewers should select 6–8 questions per session (90 min) to allow depth over breadth. Use the scoring rubric at the end of this document to calibrate feedback consistently.

Round	Title	Conducted By	Purpose
Round 1	Technical Screening	Senior AI Engineer / Tech Lead	Validate hands-on depth in AI/ML, frameworks, architecture thinking
Round 2	Architecture Deep Dive	Principal Architect / Engineering Director	Assess cloud, MLOps, system design, and coding-level decisions
Round 3	Leadership & Strategy	Director / Partner	Evaluate leadership maturity, strategic thinking, BigComp culture fit

ROUND 1 · TECHNICAL SCREENING

Hands-on AI/ML depth · Framework experience · Architecture thinking

1A | Background & Architecture Philosophy

Recommended selection: 3–4 questions · Suggested duration: 25 min

#	Question	Focus Area
Q 1	Walk me through your 12+ year journey in AI/ML engineering and how it shaped your architectural thinking.	Career Narrative
Q 2	Describe the largest enterprise-scale AI system you architected end-to-end. What was the scale, team size, and key design decisions?	System Design
Q	How do you differentiate between an AI solution that merely works	Engineering Standards

3	and one that is production-grade and enterprise-ready?	
Q 4	What does your approach to translating vague business requirements into concrete AI architecture look like? Walk through a real example.	<i>Business Alignment</i>
Q 5	How do you stay current with the rapidly evolving GenAI landscape while executing long-term architecture commitments?	<i>Technical Currency</i>
Q 6	Describe a situation where you had to push back on a stakeholder's AI vision. How did you navigate it?	<i>Stakeholder Management</i>
Q 7	What does good AI governance look like to you, and how have you embedded it into architecture decisions?	<i>Governance</i>
Q 8	How do you mentor engineers who are earlier in their AI journey without becoming a bottleneck?	<i>Leadership</i>

1B | Generative AI, LLMs & Agentic Frameworks

Recommended selection: 4–5 questions · Suggested duration: 40 min

#	Question	Focus Area
Q 1	LangGraph vs AutoGen vs CrewAI — walk me through your hands-on experience with each and when you would choose one over the others.	<i>Agentic AI Frameworks</i>
Q 2	Design a multi-agent system for financial compliance document review. What orchestration pattern would you use and why?	<i>Architecture Design</i>
Q 3	How do you handle state management, memory, and tool-calling in complex LangGraph workflows?	<i>LangGraph Deep Dive</i>
Q 4	Describe a production LLM deployment you owned. Walk through the model selection, prompt engineering, evaluation, and monitoring pipeline.	<i>LLMOps</i>
Q 5	How do you manage LLM provider failover (e.g., multi-provider fallback chains) in production systems?	<i>Resilience Patterns</i>
Q 6	What is your experience with RAG architectures? How do you handle chunking strategy, embedding choice, retrieval re-ranking, and hallucination mitigation?	<i>RAG Architecture</i>
Q 7	Walk me through fine-tuning vs in-context learning vs RAG — when do you choose each approach for enterprise use cases?	<i>Model Strategy</i>
Q 8	Describe your experience with prompt versioning, A/B testing prompts, and systematic prompt evaluation at scale.	<i>Prompt Engineering</i>

ROUND 2 · ARCHITECTURE DEEP DIVE

Cloud · MLOps/LLMOps · Distributed systems · Engineering practices

2A | Cloud Platforms, MLOps & Infrastructure

Recommended selection: 4–5 questions · Suggested duration: 40 min

#	Question	Focus Area
Q 1	Compare AWS SageMaker, Azure ML, and Vertex AI for enterprise MLOps. Where have you deployed production workloads and what were the tradeoffs?	Cloud MLOps
Q 2	Design an LLMOps pipeline from model selection through deployment, monitoring, and retraining triggers on your preferred cloud.	LLMOps Pipeline
Q 3	How do you architect for model drift detection and automatic retraining in a production ML system?	Model Monitoring
Q 4	Walk through your experience with Kubernetes-based ML serving — autoscaling, GPU node pools, spot instance strategies.	K8s / Infra
Q 5	How do you use Terraform or CloudFormation to version-control your AI infrastructure? Share a complex IaC pattern you implemented.	IaC
Q 6	Describe a CI/CD pipeline you built for model deployment using Jenkins or GitLab CI/CD. What were the quality gates?	CI/CD for AI
Q 7	How do you design for cost optimization on cloud AI workloads at enterprise scale?	FinOps
Q 8	What is your strategy for hybrid cloud and on-premise AI deployments, particularly in regulated industries?	Hybrid Architecture

2B | Engineering, APIs & Data Architecture

Recommended selection: 3–4 questions · Suggested duration: 35 min

#	Question	Focus Area
Q 1	Describe a complex PyTorch or TensorFlow model you trained, fine-tuned, or adapted for production. What optimization techniques did you apply?	Deep Learning
Q 2	How do you design Python-based AI microservices — code structure, dependency management, async patterns, and performance tuning?	Python Architecture
Q 3	Walk through the design of a GraphQL API layer over an AI backend with real-time streaming and complex query patterns.	API Design
Q 4	How do you approach distributed systems challenges in AI pipelines — consistency, partition tolerance, and backpressure handling?	Distributed Systems
Q 5	Describe your experience with vector databases (Pinecone, Weaviate, Chroma, pgvector). How do you choose between them?	Vector Storage
Q 6	How do you design multi-modal AI systems that combine text, image, and structured data inputs?	Multi-modal AI
Q 7	Walk through your experience with streaming architectures (Kafka, Redis Streams, SSE) in AI pipelines.	Streaming / Events
Q 8	How do you handle feature stores and real-time feature serving in production ML systems?	Feature Engineering

ROUND 3 · LEADERSHIP & STRATEGY*Strategic thinking · Stakeholder influence · IC leadership · Culture fit***3A | Strategy, Governance & AI Leadership***Recommended selection: 4–5 questions · Suggested duration: 40 min*

#	Question	Focus Area
Q 1	BigComp serves US-based enterprise clients in advisory and risk. How would you architect an AI Center of Excellence model for such an organization?	<i>AI CoE Strategy</i>
Q 2	Walk me through how you would assess an existing client's AI maturity and create a 12-month AI architecture roadmap.	<i>Roadmapping</i>
Q 3	How do you balance innovation velocity with enterprise governance requirements — security, compliance, audit trails?	<i>Governance vs Speed</i>
Q 4	Describe a situation where an AI project you led failed or underperformed. What was the root cause and what did you change?	<i>Failure & Learning</i>
Q 5	How do you build buy-in for architectural standards across engineering teams that may have existing opinions and habits?	<i>Change Management</i>
Q 6	What does responsible AI mean to you in the context of financial services or enterprise risk advisory?	<i>Responsible AI</i>
Q 7	How would you approach building an internal AI platform (models, APIs, guardrails, observability) for a Big Four consulting firm?	<i>Platform Thinking</i>
Q 8	As an Individual Contributor reporting to a Director, how do you influence without authority and drive cross-team architecture adoption?	<i>IC Leadership</i>

3B | Scenario-Based Design Challenges*Recommended selection: 2–3 questions · Suggested duration: 30 min · Whiteboard or verbal walk-through encouraged*

#	Question	Focus Area
Q 1	A client wants to use an LLM to auto-generate regulatory filings. Design the end-to-end architecture including human-in-the-loop, auditability, and rollback.	<i>Scenario Design</i>
Q 2	You are asked to reduce LLM inference costs by 60% without degrading quality. Walk through your strategy.	<i>Cost Optimization</i>
Q 3	A GenAI application is leaking sensitive PII in outputs. How do you triage, contain, and prevent recurrence architecturally?	<i>Security Incident</i>
Q 4	Design a multi-tenant AI SaaS system where each client's data must be strictly isolated but infrastructure is shared.	<i>Multi-tenancy</i>
Q	What is your approach to evaluating a third-party AI vendor's claims	<i>Vendor Evaluation</i>

5 — model cards, benchmarks, security posture?

Evaluation & Scoring Rubric

Score each dimension on a 1–3 scale. Aggregate across rounds before making a hiring decision. Any single dimension scored 1 by two or more interviewers should trigger a debrief discussion.

Dimension	1 – Below Bar	2 – Meets Bar	3 – Exceeds Bar
Technical Depth	Surface-level answers; no hands-on evidence	Solid fundamentals; demonstrates production experience	Expert-level; cites edge cases, failure modes, tradeoffs
Architecture Thinking	Narrow point solutions; no systems view	Designs scalable systems; considers non-functional reqs	Anticipates future needs; clear design philosophy
LLM/GenAI Maturity	Aware of tools but limited practical application	Has shipped GenAI features; familiar with LLMops	Owned full LLM product lifecycle; aware of frontier research
Communication	Struggles to explain tradeoffs to non-engineers	Clear and structured; adapts to audience	Compelling storyteller; influences through clarity
Leadership & Mentoring	Focuses only on individual contribution	Coaches peers; drives team standards	Builds engineering culture; grows others proactively

Red Flags to Watch For

- Cannot articulate tradeoffs — only describes what they built, not why alternative approaches were rejected
- LLM experience confined to prompt engineering with no systems-level ownership
- No evidence of having led or influenced architecture across teams
- Vague answers on security, governance, or compliance — especially concerning for US financial clients
- Cannot discuss failure modes or production incidents — suggests limited real-world exposure

Strong Signal Indicators

- Unprompted mentions of evaluation frameworks, observability, and model lifecycle management
- Cites specific production scale metrics (QPS, latency p99, cost per inference, token budget)
- Demonstrates knowledge of BigComp-relevant domains: risk advisory, audit, regulatory compliance
- Shows intellectual humility — acknowledges what they don't know and how they would learn it
- References open-source contributions, patents, or published work as evidence of craft

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